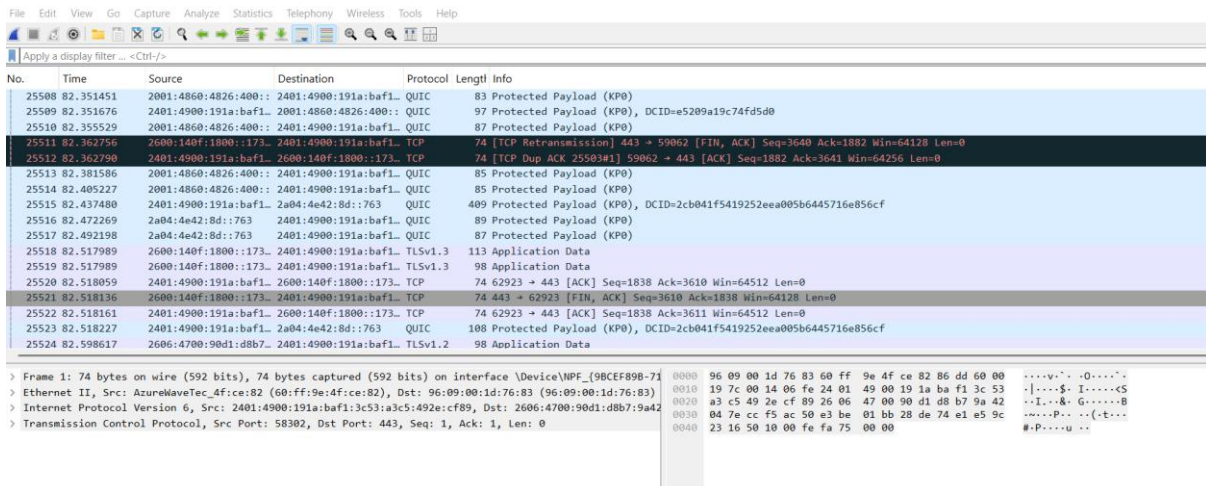


Software and Hardware Tools

Task 1.

Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.



Protocol s

TLS Encrypts web traffic (HTTPS)

TCP Reliable data transmission

DNS Resolves domain names into IP addresses

QUIC Transport-layer protocol running over UDP, primarily associated with HTTP/3.

Task 2.

Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

```

C:\Users\sketan>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:8a00::17c1:a553] with 32 bytes of data:
Reply from 2600:140f:8a00::17c1:a553: time=97ms
Reply from 2600:140f:8a00::17c1:a553: time=111ms
Reply from 2600:140f:8a00::17c1:a553: time=140ms
Reply from 2600:140f:8a00::17c1:a553: time=359ms

Ping statistics for 2600:140f:8a00::17c1:a553:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 97ms, Maximum = 359ms, Average = 176ms

C:\Users\sketan>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=97ms
Reply from 2a04:4e42:31::810: time=108ms
Reply from 2a04:4e42:31::810: time=121ms
Reply from 2a04:4e42:31::810: time=139ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 97ms, Maximum = 139ms, Average = 116ms

```

Website	Average Response Time
---------	-----------------------

Zomato	176 ms
--------	--------

Spotify	116 ms
---------	--------

Task 3.

Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Process Name	Memory Usage
--------------	--------------

Google chrome	1183. MB
---------------	----------

Microsoft edge	20 .2 MB
----------------	----------

Wireshark	17.02 MB
-----------	----------

Task 4.

Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.

IP Address	Device Name
192.168.56.1	PSLaptop
192.168.1.105	Nothing Phone (1)

Task 5.

Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tools Used in Network Troubleshooting

Tool	When It Is Used
Cable Tester	Checks whether Ethernet cables are wired correctly and detects broken wires.
Crimping Tool	Attaches RJ-45 connectors to Ethernet cables when making or repairing network cables.
Digital Multimeter	Measures voltage, continuity, and resistance when troubleshooting powered network equipment or cables.